

Charger Specifications for a One-Cell Lithium Iron Phosphate Battery System

The charger for this project must be selected to match the chosen **one-cell lithium iron phosphate battery** and the **one-cell lithium iron phosphate protection circuit module**. The charger must be compatible with the battery chemistry, the battery voltage limits, and the intended charging setup.

Because charging will **not** occur while the system is running, the charger does not need load-sharing or power-path features. A simpler charger is acceptable as long as it charges the battery correctly and safely.

Required charger characteristics

1) The charger must be for a one-cell lithium iron phosphate battery

The charger must be specifically designed for a **one-cell lithium iron phosphate battery** (often written as **1S LiFePO4**).

This is the most important requirement because lithium iron phosphate batteries use a different charging voltage than regular lithium-ion or lithium-polymer batteries. A charger meant for a regular 4.2-volt lithium-ion cell is not appropriate for this battery chemistry.

2) The charger must use a constant-current / constant-voltage charging method

The charger must use a **constant-current / constant-voltage** charging profile.

This is the correct charging approach for lithium iron phosphate batteries and is also consistent with the selected protection circuit module specification, which lists a lithium iron phosphate charging voltage using a constant-current / constant-voltage method.

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3) The charger's constant-voltage target must match the battery chemistry

The charger must regulate to the correct **final charging voltage for one-cell lithium iron phosphate** operation.

For this project, the selected battery datasheet lists a **maximum battery voltage of 3.65 volts**, and the selected protection circuit module documentation lists **3.6 volts constant-current / constant-voltage charging**.

This means the charger must be a lithium iron phosphate charger with a final voltage in the proper lithium iron phosphate range (for example, a charger marketed as **3.6-volt lithium iron phosphate** charging).

4) The charger current must be within the battery's allowed charging current

The charger's output current must be selected so it is safely within the battery's charging limits.

The selected battery datasheet allows a **standard charge rate up to 5 amperes (1C max)**.

This means the charger current can be lower than that maximum and still be fully acceptable.

For this prototype, a **moderate charging current** is usually the better choice because it reduces heat and simplifies charging hardware integration.

A practical target is:

- **approximately 0.5 ampere to 2 amperes charging current** for prototype use

This range is comfortably below the battery's listed maximum charging rate and is usually easier to manage thermally.

5) The charger current must also be compatible with the protection circuit module

The charger must be compatible with the selected protection circuit module, which is a **one-cell lithium iron phosphate protection board**.

This means the charging current chosen for the charger must not exceed what the protection circuit module can safely pass during charging. The protection circuit module is a backup safety device and should not be forced into unnecessary trips during normal charging.

6) The charger must be electrically compatible with the intended input source

The charger must accept the input source that will be used in the project (for example, a 5-volt external source or USB-based charging source, if that is the intended charging method).

This means the charger datasheet or listing must clearly state:

- the required input voltage range
- the expected input connection type (for example, connector or solder pads)

This requirement ensures the charger can actually be powered in the intended prototype setup.

Recommended charger characteristics (strongly preferred)

7) The charger should have clear charge termination behavior

The charger should clearly specify how charging ends or transitions when the battery is full.

This is important for repeatable charging, battery health, and safety. A charger with documented charge termination behavior is easier to trust and easier to test.

8) The charger should include thermal regulation or thermal protection

The charger should include thermal protection or thermal regulation, especially if charging occurs in a compact enclosure or wearable-related packaging.

This helps reduce charging risk and improves reliability, even if the selected charging current is moderate.

9) The charger should include charging status indication

A charger with charging status indication is strongly preferred.

Useful status outputs include:

- charging in progress
- charge complete

These may be indicator lights or status pins. Status indication makes debugging and testing much easier during prototyping.

10) The charger should have clear documentation and wiring information

The charger should have a real specification sheet or clearly documented product listing.

The documentation should include:

- one-cell lithium iron phosphate compatibility
- final charging voltage
- charging current (fixed or adjustable)
- input voltage range
- wiring / terminal labels
- board dimensions (if a module is used)

A vague listing that only says “lithium charger board” without chemistry details should not be used.

Features that are not required for this project

Because charging will not happen while the system is operating, the charger does **not** need:

- power-path management
- load sharing
- simultaneous battery charging and system powering features

This simplifies charger selection and makes a basic lithium iron phosphate charger a suitable option.

Final summary of charger requirements for this project

The charger for this project must be:

- a **one-cell lithium iron phosphate charger**
- a **constant-current / constant-voltage charger**
- set for the correct **lithium iron phosphate final charging voltage** (consistent with the selected battery and protection circuit module)
- set to a charging current that is within the battery’s allowed charge current (battery allows up to 5 amperes standard charge)

- compatible with the protection circuit module and intended charging input source

A charger in this category will correctly support the selected battery and protection circuit module for the project's non-charging-while-running design.

Chosen Charger: **Smart Charger (0.7 A) for 3.2V LiFePO4 Battery Pack, 100-240VAC, CE listed -**